COURSE SYLLABUS

Weekly Subjects and Related Preparation Studies

WEEK	SUBJECT	DATE	
1	INTRODUCTION BASIC DEFINITIONS (TERMS)	05.10.2020 08.10.2020	Monday Thursday
2	ROAD USER CHARECTERISTICS VEHICLE MOVEMENTS	12.10.2020 15.10.2020	Monday Thursday
3	TRAFFIC FLOW AND CAPACITY ROADWAY CASSIFICATION AND GEOMETRIC STANDARDS	19.10.2020 22.10.2020	Monday Thursday
4	HIGHWAY ROUTE STUDY **NATIONAL HOLIDAY**	26.10.2020 29.10.2020	Monday Thursday
5	HORIZANTAL CURVE AND SUPERELEVATION (Project sheet distribution) HORIZANTAL CURVE AND SUPERELEVATION	02.11.2020 05.11.2020	Monday Thursday
6	**APPLICATION (VEHICLE MOVEMENTS)** **ASSIGNMENT - APPLICATION (TRAFFIC)**	09.11.2020 12.11.2020	Monday Thursday
7	**APPLICATION (HORIZANTAL CURVE)** **ASSIGNMENT-APPLICATION (HORIZANTAL CURVE)**	16.11.2020 19.11.2020	Monday Thursday
8	**MID TERM EXAM**	23.11.2020 26.11.2020	Monday Thursday
9	TRANSITION CURVES- ASSIGNMENT (HAND IN) TRANSITION CURVES– ASSIGNMENT (RETURN)	30.11.2020 03.12.2020	Monday Thursday
10	VERTICAL ALIGNMENT AND VERTICAL CURVE VERTICAL ALIGNMENT AND VERTICAL CURVE	07.12.2020 10.12.2020	Monday Thursday
11	**APPLICATION (TRANSITION CURVES)** **ASSIGNMENT**	14.12.2020 17.12.2020	Monday Thursday
12	**APPLICATION (VERTICAL CURVE)** **APPLICATION (VERTICAL AND HORIZANTAL CURVE)**	21.12.2020 24.12.2020	Monday Thursday
13	PLANNING OF INTERSECTION AND URBAN ROADS PROJECTS DELIVERY DUE DATE (December 30, 2020)	28.12.2020 31.12.2020	Monday Thursday
14	DRAINAGE IN HIGHWAY **MAKEUP EXAM**	04.01.2020 07.01.2020	Monday Thursday

CHARACTERISTICS OF HIGHWAY USERS

The characteristics of **human** and **vehicle** using the highway have to be known. Two main characteristics vehicle and highway are designed, built and controlled by the human.



Normal Phsical Charectiristics

Visual characteristics

Visual Angle: The angle on which a person can see clearly when holding the head high and looking forward. When the head and the eyes are remained constant, visual angle is 5º on each side of the vertical plane, making a total of 10º. There's also a horizontal plane and together they form a visual cone. **Clear visual angle** can scan each side with 2,5º.

Visibility Distance : The distance, which a person is able to see clearly on forward direction

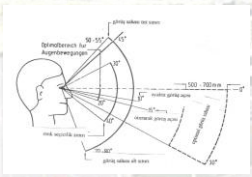
Colorblindness: Prevents recognizing red and green on traffic signals is a serious problem and those people are legally forbidden from driving.

Dazzle: Short term visual ability loss under sudden and intense light
- entering a dark environment from light 6 sec.
- entering a light environment from dark 3 sec.

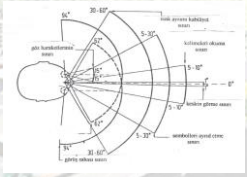
CHARACTERISTICS OF HIGHWAY USERS

The characteristics of **human** and **vehicle** using the highway have to be known. Two main characteristics vehicle and highway are designed, built and controlled by the human.

Vertical Visual Angle



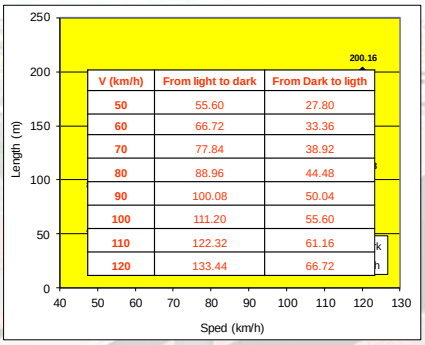
Horizantal Visual Angle



Speed (km/h)	90	100	110	120
Visual angle (derece)	25	20	16	12
Visibility distance (m)	500	600	700	750

CHARACTERISTICS OF HIGHWAY USERS

Driving Length for different speed under dazzle



V (km/h)	From light to dark	From Dark to ligth
50	55.60	27.80
60	66.72	33.36
70	77.84	38.92
80	88.96	44.48
90	100.08	50.04
100	111.20	55.60
110	122.32	61.16
120	133.44	66.72

CHARACTERISTICS OF HIGHWAY USERS

Reading Time Allowance to Traffic Messages

Dudek (1990) Formula;

"Research...has indicated that a minimum exposure time of **one second per short word** (four to eight characters) (exclusive of prepositions and other similar connectors) **or two seconds per unit of information,** whichever is largest, should be used for unfamiliar drivers. On a sign having **12 to 16 characters per line, this minimum exposure time will be two seconds per line**." "Exposure time" can also be interpreted as "reading time" and so used in estimating how long drivers will take to read and comprehend a sign with a given message

CHARACTERISTICS OF HIGHWAY USERS

Suppose a sign reads:

Traffic Conditions
Next 2 Miles
Disabled Vehicle on I-77
Use I-77 Bypass Next Exit

How long does it take to read that message?

CHARACTERISTICS OF HIGHWAY USERS

Normal Phsical Charectiristics



Distance estimation characteristic
It is important for **a driver to estimate the distance of a vehicle in front or behind**, and also **the distance of a pedestrian crossing the street.** Many traffic accidents occur because of distance estimation errors. The age, experience and education of the driver is important on distance estimation.

Auditory characteristics
Drivers and pedestrians should have at least a normal auditory level to react to the warnings and announcements made vocally on highways.

Perception-reaction characteristics
The total time needed for a driver or a pedestrian to see **a danger on the highway**, **percieve it**, **decide what to do** and **apply** his/her reaction is named as the perception-reaction time.

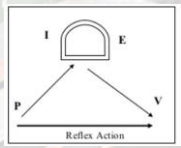
CHARACTERISTICS OF HIGHWAY USERS

Perception-reaction characteristics

- **PIEV THEORY**
  - PERCEPTION
  - INTELLECTION
  - EMOTION
  - VOLITION

CHARACTERISTICS OF HIGHWAY USERS

- PERCEPTION: Time required to perceive an object
- INTELLECTION: Time required for understanding the situation.
- EMOTION: Time elapsed during emotional sensations and disturbances such as fear, anger or other feelings.
- VOLITION: Time taken for final action



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CHARACTERISTICS OF HIGHWAY USERS

Perception-reaction time

USA	2.5 sec.
JAPAN	
CANADA	
SOUTH AFRICA	

FRANCE	2 sec
GERMANY	
ENGLAND	

In literature, it is suggested as;
1.5 sec. for urban highways and
2.0 sec. for rural highways.

CHARACTERISTICS OF HIGHWAY USERS

Perceaption-reaction time for different reliability level

%50 reliability	1.260 sec.
%85 reliability	1.665 sec.
%99 reliability	2.280 sec.

Perceaption-reaction time should be increased
0.5 sec
if driver is not warned with a traffic sign or any messages.

Speed (km/h)	Length (m)
50	6.94
90	12.50
100	13.89
120	16.67
130	18.06

CHARACTERISTICS OF HIGHWAY USERS

Speed and Perceaption-reaction time releationship

Perceaption-reaction time (sec.) \ Speed (km/h)	50	60	70	80	90	100	110	120
0.5	**6.9**	**8.3**	**9.7**	**11.1**	**12.5**	**13.9**	**15.3**	**16.7**
1.0	13.9	16.7	19.4	22.2	25.0	27.8	30.6	33.3
1.5	**20.8**	**25.0**	**29.2**	**33.3**	**37.5**	**41.7**	**45.8**	**50.0**
2.0	**27.8**	**33.3**	**38.9**	**44.4**	**50.0**	**55.6**	**61.1**	**66.7**
2.5	34.7	41.7	48.6	55.6	62.5	69.4	76.4	83.3

CHARACTERISTICS OF HIGHWAY USERS

Temporary physical characteristics

The effects of these characteristics can be seen on a short or long term and are results of **fatigue**, **disease** and usage of pleasure-inducing substances (**Driving under the influence** (DUI))



Mental characteristics

They can be summarized as **intelligence**, **knowledge**, **good manners**, **experience**, **reading** and **foreign language knowledge**.

Psychological characteristics

Some of them can be listed as, **gathering attention**, **corresponding with the rules**, **being patient and calm**, **tendency to high speed and accidents**.





CHARACTERISTICS OF HIGHWAY USERS

Age of driver

- Problems of younger drivers
  - Difficulty in judging speed, distance and reaction time
  - Tend to concentrate on near objects
  - Missing the important information, because its relevance is not understood
  - Having difficulty in integrating information
  - Under-estimating the risk of accident involvement
  - Making less effective driving decisions

CHARACTERISTICS OF HIGHWAY USERS

Age of driver

- Problems of older drivers
  - Difficulties in rapid decision making
    - At intersections
  - Take time to absorb traffic control information
  - Difficulty at night
    - Lower light level
    - Headlight glare
  - Readily fatigued

CHARACTERISTICS OF HIGHWAY USERS

Pedestrian Walking Speed

Pedestrian-vehicle accidents may occur on traffic especially when pedestrians are crossing a street. For this reason it is important to know the walking speed of pedestrians. It is especially used on determining the green light time for pedestrians of traffic lights.

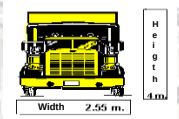
Age of the Pedestrian	Pedestrian walking speed (m/sec)			
	Under heavy traffic conditions		Under light traffic conditions	
	Normal	Fast	Normal	Fast
0-7	1,13	2,40	1,29	2,33
7-15	1,34	2,40	1,16	2,91
15-25	1,34	2,15	1,24	2,33
>25, male	2,40	2,40	2,33	2,33
>25, female	1,20	2,15	1,16	2,33

Vehicle Characteristics

Vehicle: According to the Highway Traffic Law 2918, cars which carry people, animal or freight on highways. Highway vehicles can be categorized into two: as motorized vehicles and unmotorized vehicles.

Vehicle dimensions and weight (Karayolları Trafik Yönetmeliği 128. maddesi)

Dimension of Vehicle



Width : Affect width of traffic lanes, shoulders & parking facilities .

Height: Affects the clearance of overhead structures.

- Clearance under over bridges, under bridges, electric & other service lines.
- Height of driver seat –provide clear vision of the road to his front & sides.
- Height of headlight- affects sight distances at valley curves.

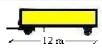
Vehicle Characteristics

The length of vehicle (Karayolları Trafik Yönetmeliği)

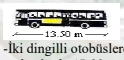
c- Azami uzunluklar aşağıda verilmiştir. Verilen uzunluklar taşıt tipine göre belirlenmiştir ve daha uzun yapılamaz.



-Otobüs dışındaki diğer motorlu araçlarda: 12.00 m.



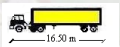
-Römorklarda ; 12.00 m.



-İki dingilli otobüslerde : 13.50 m. otobüslerde:15.00 m.



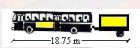
-İkiden çok dingilli



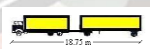
-Yarı römorklu araçlarda: 16.50 m. 18.75 m.



-Mafsallı (Körüklü) otobüslerde:



-Römorklu otobüslerde : 18.75 m.



-Römorklu araçlarda : 18.75 m.

Vehicle Characteristics

Length: Important factor in design of Horizontal alignment

- Length affects
  - Safe overtaking distance
  - Capacity of road
  - Extrawidening
  - minimum turning radius
  - Parking Facilities
  - Design of valley curves.

Vehicle Characteristics

Araçların Boyutları



BOYUTLAR	
Dingil (mm)	2540
Ön/Arka iz genişliği (mm)	1470/1440
Uzunluk/Genişlik/Yükseklik * (m)	4,39/1,74/1,43



BOYUTLAR	
Uzunluk (mm)	4460
Genişlik (mm)	1715
Aks mesafesi (mm)	2625

Vehicle Characteristics

Araçların Boyutları



BOYUTLAR	
Uzunluk	11980 mm
Genişlik	2500 mm
Yükseklik	3977 mm
Dönüş dairesi çapı	21292 mm



BOYUTLAR	
Uzunluk	13850 mm
Genişlik	2550 mm
Yükseklik	3710 mm



BOYUTLAR	
Uzunluk	11825 mm
Genişlik	2500 mm
Yükseklik	2865 mm



BOYUTLAR	
Uzunluk	17825 mm
Genişlik	2500 mm
Yükseklik	2980 mm

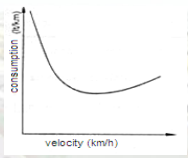
Vehicle Characteristics

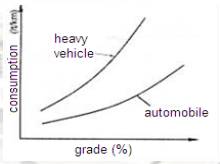
Minimum vehicle turning radius: Minimum design turning radius is mostly used for very slow vehicles (usually with a speed less than 15 km/h). Turning radius is around 7 m for small vehicles, 13 m for trucks, 15 m for semi-tractors for 90º turn.

Acceleration and deceleration of vehicles: Acceleration values vary between 0,5-3,0 m/sec² for various types of vehicles. The deceleration value for a vehicle moving with a velocity of 100 km/h is around 1 m/sec² .

Operating costs of motorized vehicles: The operation costs of vehicles can be categorized as fixed and variable expense items. Variable expenses emerge as vehicle starts to move. Fixed expenses are independent of vehicle movements.

Vehicle Characteristics





For example, for a highway section of 2 km with %5 grade, operating costs of trucks are approximately

operating costs of trucks

$L_{eq,\ truck}$= 2*(2,25)= 4,5 km of straigth highway

operating costs of ligth vehicle

$L_{eq,\ truck}$= 2*(1,14)= 2,285 km of straigth highway